

Student: Jonatan Rabjerg**Instructor:** Andreas Basse-O'Connor**Assignment:** MyLab Math - Uge 2**Date:** 1/15/24**Course:** Calculus Beta

Use implicit differentiation to find $\frac{dy}{dx}$.

$$x^7 - 70xy + y^7 = 1$$

Differentiate both sides of the equation with respect to x , treating y as a differentiable function of x .

$$x^7 - 70xy + y^7 = 1$$

$$\frac{d}{dx}(x^7) - \frac{d}{dx}(70xy) + \frac{d}{dx}(y^7) = \frac{d}{dx}(1)$$

$$7x^6 - 70\left(x\frac{dy}{dx} + y\frac{dx}{dx}\right) + 7y^6\frac{dy}{dx} = 0$$

Collect the terms with $\frac{dy}{dx}$ on one side of the equation and solve for $\frac{dy}{dx}$.

$$7x^6 - 70\left(x\frac{dy}{dx} + y\frac{dx}{dx}\right) + 7y^6\frac{dy}{dx} = 0$$

$$7x^6 - 70x\frac{dy}{dx} - 70y + 7y^6\frac{dy}{dx} = 0$$

$$7x^6 - 70y = \frac{dy}{dx}(70x - 7y^6)$$

$$\frac{7x^6 - 70y}{70x - 7y^6} = \frac{dy}{dx}$$

$$\frac{x^6 - 10y}{10x - y^6} = \frac{dy}{dx}$$

Therefore, the derivative is $\frac{dy}{dx} = \frac{x^6 - 10y}{10x - y^6}$.